

Budget Justification

NSF CISE CORE (IIS) - CopernicusAI Knowledge Engine

Principal Investigator: Gary Welz

Project Period: 36 months (3 years)

Note on Budget Revisions

This budget has been revised per CUNY Graduate Center Sponsored Research: (1) Fringe benefits added using GC rates (PI summer salary 25.05%; part-time personnel 8% + 0.895% MTA); (2) Indirect cost rate set to 54% (research) applied to MTDC; (3) Materials & Supplies set to \$15,000 total to align with budget; (4) All justification totals match the budget.

A. Senior Personnel (\$180,000 total)

Principal Investigator (Gary Welz)

- Year 1: 6 months @ \$10,000/month = \$60,000
- Year 2: 6 months @ \$10,000/month = \$60,000
- Year 3: 6 months @ \$10,000/month = \$60,000
- **Total: \$180,000**

Justification: The PI will lead all research activities including interface design, metadata schema development, reliability mechanism implementation, and evaluation studies. Six months per year effort reflects the technical and research-intensive nature of the work.

B. Other Personnel (\$78,000 total)

Research Assistant / Software Developer: \$60,000 total (Year 1–3: \$20,000/year at 4 months @ \$5,000/month)

Student User Testers (Graduate): \$18,000 total (Year 1–3: \$6,000/year for 2 graduate students @ \$15/hour, 10 hrs/week, 20 weeks)

Justification: Research assistant provides development capacity for metadata pipelines, cross-modal linking, and evaluation infrastructure. Student testers support usability testing, collaborative validation, and real-world usage data.

C. Fringe Benefits (\$52,029 total)

Rates per CUNY Graduate Center: **PI summer salary 25.05%; part-time personnel 8% + 0.895% MTA.**

- **PI:** 25.05% x \$60,000/year = \$15,030/year = **\$45,090 total**
- **Research Assistant:** 8.895% x \$20,000/year = \$1,779/year = **\$5,337 total**
- **Student Testers:** 8.895% x \$6,000/year = \$534/year = **\$1,602 total**

Total Fringe: \$52,029 (\$17,343 per year)

D. Equipment (\$12,000 total)

- **Year 1:** \$8,000 (development laptop, tools, software; monitoring/analytics; cloud credits)
- **Year 2:** \$2,000 (software renewals; cloud credits)
- **Year 3:** \$2,000 (software renewals; cloud credits)

Justification: Supports development, integration, and testing. Cloud credits provide buffer during intensive development and user studies.

E. Travel (\$16,000 total)

- **Year 1:** \$4,000 (NSF PI/CISE meeting; AI/research tools conference)
- **Year 2:** \$6,000 (NSF meeting; conference presentation; user study site visits)
- **Year 3:** \$6,000 (NSF meeting; final results conference; user study site visits)

Justification: Travel for NSF meetings, conferences, and user study site visits.

F. Participant Support (\$0)

Not applicable.

G. Other Direct Costs (\$115,000 total)

Cloud Infrastructure & Services: \$78,000 total

- Year 1: \$25,000 (GCP, third-party AI, Vercel, network)
- Year 2: \$28,000 (increased usage for user studies and scaling)
- Year 3: \$25,000

Consultants/Subcontracts: \$15,000 total (Year 1: \$5,000; Year 2: \$5,000; Year 3: \$5,000)

Materials & Supplies: \$15,000 total (Year 1: \$5,000; Year 2: \$5,000; Year 3: \$5,000)

Publication & Documentation: \$4,000 total (Year 1: \$1,000; Year 2: \$1,000; Year 3: \$2,000)

Other: \$3,000 total (Year 1: \$1,000; Year 2: \$1,000; Year 3: \$1,000)

*Year 1 Other Direct: \$37,000 | Year 2: \$40,000 | Year 3: \$38,000 | **Total: \$115,000***

Justification: Cloud infrastructure supports production-scale development and multi-model LLM services. Consultants provide metadata design, QA, and evaluation expertise. Materials and publication support documentation and dissemination.

H. Indirect Costs (\$238,155 total)

Indirect Cost Rate: 54% (CUNY Graduate Center negotiated rate for research) applied to modified total direct costs (MTDC). MTDC excludes equipment.

- Year 1: $54\% \times (\$152,343 - \$8,000) = 54\% \times \$144,343 = \mathbf{\$77,945}$
- Year 2: $54\% \times (\$151,343 - \$2,000) = 54\% \times \$149,343 = \mathbf{\$80,645}$
- Year 3: $54\% \times (\$149,343 - \$2,000) = 54\% \times \$147,343 = \mathbf{\$79,565}$
- **Total: \$238,155**

Justification: CUNY Graduate Center negotiated indirect cost rate for research (54%).

Budget Justification Summary

This budget supports a 3-year research project to develop and evaluate an AI-enabled assisted research platform. The budget allocates:

- **35% to Senior Personnel:** PI effort (6 months/year)
- **15% to Other Personnel:** Research assistant and graduate student user testers
- **10% to Fringe Benefits:** CUNY GC rates (PI summer 25.05%; part-time 8% + 0.895% MTA)
- **22% to Other Direct Costs:** Cloud infrastructure, consultants, materials (\$15,000), publication, other
- **34% to Indirect Costs:** CUNY GC research rate (54% of MTDC)

Totals: Total Direct Costs \$453,029; Total Indirect Costs \$238,155; **Total Request \$691,184.**

Cost Sharing

No cost sharing is proposed.

Budget Prepared By: Gary Welz, Principal Investigator

Total Direct Costs Requested: \$453,029

Total Indirect Costs: \$238,155 (54% of MTDC; CUNY GC research rate)

Total Request: \$691,184

Fringe and indirect rates per CUNY Graduate Center Office of Sponsored Research. Materials & Supplies total \$15,000.